Evolution in simulation (20 robots, windy corridor)

Random rules

ABCD ~ U[-5, 5]
880 values,
shared by all robots

Stage 1

walk upwind
100 generations
$f = 16\, x_{avg}$

Stage 2

+ battery and collisions
200 generations
$f = 16\, x_{avg} + B_{avg,\,end}/5$
$- (t_{col} + 3\, t_{wcol})/250$

Each stage, every generation (CMA-ES)

Sample

$\lambda = 30$
candidate
rule sets

Simulate

3 runs each,
20 robots,
random starts

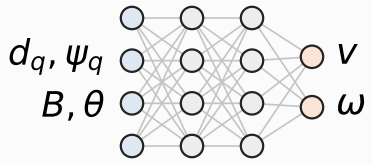
Score

median
fitness of
the 3 runs

Update

move search
distribution;
keep best-ever

next generation



Each robot: 10-10-10-2 network,
random start weights; every step:

$$\delta w_{ij} = \mu(A_{ij} n_i n_j + B_{ij} n_i + C_{ij} n_j + D_{ij})$$

Evaluation

Simulation

vs. hand-designed
baseline (Ferrante et al.),
same start positions,
1 to 20 robots

Real robots

7 Thymio-II robots

Lighthouse positions,
shared by radio,
give the same 10 inputs

wind and battery
computed on board